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| 4(a)     | <i>four from:</i><br>1 spontaneous / random, mutation ;<br>2 produces an allele coding for insect-deterring chemicals ;<br>3 (grazing by) insects, is selection pressure ;<br>4 plants with, alleles / mutations, have selective advantage / were selected for / <b>ora</b> ;<br>5 (plants with alleles / mutations) survived / reproduced / produced more seeds / <b>ora</b> ;<br>6 passed on this allele (to offspring) ;<br>7 this allele increases in frequency ; | <b>4</b> |
| 4(b)(i)  | <u>directional</u> ;                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>1</b> |
| 4(b)(ii) | <u>stabilising</u> ;                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>1</b> |
| 4(c)     | (genetic drift) occurs due to chance events / <b>ora</b> for natural selection ;<br>(genetic drift) shows no trend in results / <b>ora</b> for natural selection ;                                                                                                                                                                                                                                                                                                    | <b>2</b> |